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4 / 4 submitted

Q1. Electric Battery Manufacturing

Score: 25.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    int monthlyProduction;
    scanf("%d", &monthlyProduction);

    if(monthlyProduction > 5000)
        printf("Yes Incentive\n");
    else
        printf("No Incentive\n");

    if(monthlyProduction < 4000)
        printf("Yes Inspection\n");
    else
        printf("No Inspection\n");

    return 0;
}
```

INPUT

3500 42000 3500

OUTPUT

No Incentive
Yes Inspection

Q2. Maintenance

Score: 25.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    int load, maintenance;
    float percentage;
    int available;

    scanf("%d %d", &load, &maintenance);

    if (maintenance == 1) {
        return 0;
    }

    if (load > 500) {
        printf("Grid Failure\n");
        return 0;
    }

    percentage = (load / 500.0) * 100;
    available = 500 - load;

    printf("Load Percentage = %.0f%%\n", percentage);
    printf("Available Capacity = %d MW\n", available);

    if (percentage ≤ 60)
        printf("Status = Normal\n");
    else if (percentage ≤ 85)
        printf("Status = High Load\n");
    else
        printf("Status = Critical\n");

    return 0;
}
```

INPUT

250

OUTPUT

Load Percentage = 50%
Available Capacity = 250 MW
Status = Normal

Q3. Medical

Score: 25.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    int totalStock = 250000;
    int doses;
    int hospitalStatus;    // 1 = OPEN, 0 = CLOSED

    scanf("%d %d", &doses, &hospitalStatus);

    if (hospitalStatus == 0)
        return 0;

    totalStock = totalStock - doses;

    if (doses ≥ 10000)
        printf("Hospital Status = Sufficient\n");
    else if (doses ≥ 8000)
        printf("Hospital Status = Minimum Requirement\n");
    else
        printf("Hospital Status = Emergency Supply Required\n");

    printf("Remaining Stock = %d doses\n", totalStock);

    return 0;
}
```

INPUT

12000

OUTPUT

Hospital Status = Sufficient
Remaining Stock = 238000 doses

Q4. Storage

Score: 25.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    int used, total;
    int remaining, extraStorage, extraCharge;
    char plan[20];

    scanf("%d %s", &used, plan);

    if(strcmp(plan, "Basic") == 0)
        total = 100;
    else if(strcmp(plan, "Standard") == 0)
        total = 500;
    else if(strcmp(plan, "Enterprise") == 0)
        total = 2000;
    else
    {
        printf("Invalid Plan");
        return 0;
    }

    if(used > total)
    {
        remaining = 0;
        extraStorage = used - total;
    }
    else
    {
        remaining = total - used;
        extraStorage = 0;
    }

    extraCharge = extraStorage * 4;

    /* Case 1 : Normal */
    if(extraStorage == 0 && (used * 100 / total) < 95)
    {
        printf("Total Storage = %d GB\n", total);
        printf("Remaining = %d GB\n", remaining);
        printf("Extra Storage = 0 GB\n");
        printf("Extra Charge = Rs. 0\n");
        printf("Status = Normal");
    }

    /* Case 2 : Extra Storage */
    else if(extraStorage > 0)
    {
        printf("Remaining = %d GB\n", remaining);
        printf("Extra Storage = %d GB\n", extraStorage);
        printf("Extra Charge = Rs.%d\n", extraCharge);
        printf("Status = Extra Storage Used");
    }

    /* Case 3 : Warning */
    else
    {
        printf("Remaining = %d GB\n", remaining);
        printf("Extra Charge = Rs. 0\n");
        printf("Warning Generated\n");
        printf("Status = Storage Nearly Full");
    }
}
```

```
    return 0;  
}
```

INPUT

120 Basic

OUTPUT

Remaining = 0 GB
Extra Storage = 20 GB
Extra Charge = Rs.80
Status = Extra Storage Used